Database Rollback using a Restore Point

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1 Database Rollback

Valid for databases with installed OCC Admin package only.

During an application upgrade or migration of a TC Oracle database it may be necessary to roll back the content of the database to a former state.
This can be done by

- taking a dump of the database before start of the migration. If necessary this dump can be used to set the database back to the date time before the migration started.
- setting a restore point before starting the migration. Then the database can be rolled back to the time the restore point set or to a dedicated SCN or time between setting the restore point and the actual time (point of timee recovery)

This describes the rollback of a database based on a restore point using the OCC AdminPackage (P_ADM).

1.1 Prerequisite

To use RESTORE POINTs the database has to be running in ARCHIVELOG mode (**mandatory**).
Restore point administration provides some procedures to handle guaranteed restore points.
Before any procedure can be used an **OCC DBA** has to **pre-configure the restore point** among other things (**mandatory**).
Very important is the sizing of the so called flashback area. It's limiting the volume of changes to the database.
By using the OCC Admin package only one restore point can be active at the same time.
So

- To use restore points the function has to be activated by OCC.
- If activated the OCC-customer (i. e. CI/NEM) may create, drop, flashback to and simulate creation of guaranteed restore points.
- Restore Point have to be configured by OCC before they can be used..
- Concurrently only one restore point can be created in the database but many restore points can be preconfigured.
- A version of p_adm package has to be installed in database that covers the needed functionality.

1.2 Usage of a restore point

The normal way to handle restore points is the following:

1. Check preconfigured (unused) Restore Points with an valid time frame to activation.
   Activation can be done in time frame VALID_FROM to VALID_TO.
2. Create a restore point by procedure CreateRP
3. Perform application maintenance
4. If maintenance was successful
   then drop restore point by procedure DropRP
   else if maintenance was not successful: Flashback to restore point. Then decide to drop the restore point or to start with (2) again.

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Without dropping a restore point application will no longer be able to do insert/update/deletes when the flashback area is full. So it's important to drop the restore point when the maintenance is over.

1.3 Check preconfigured Restore Points

All preconfigured Restore Points provided by OCC-DBAs that were used or can be used by customer are contained in table ADM_RESTORE_POINTS
To check which Restore Points are available and unused check this table.

Login in database via sqlplus as **ADMUSER** and execute a select on this table.

Example:

```
d:\Puffer\oracle>sqlplus admuser@te4401.world

SQL*Plus: Release 11.2.0.2.0 Production on Do Feb 26 13:56:07 2015

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Kennwort eingeben:

Verbunden mit:
Oracle Database 11g Enterprise Edition Release 11.2.0.4.0 - 64bit Production
 set lines 132 pages 9999
 column VALID_FROM format A25
 column VALID_TO format A25
  column SCN format 999999999999999
 set serveroutput on
 select RP_NO, VALID_FROM, VALID_TO, SCN, STATUS_CREATED, STATUS_DROPPED, STATUS_FLASHBACK  from
adm.adm_restore_points;

     RP_NO VALID_FROM                VALID_TO                          SCN STATUS_CREATED STATUS_DROPPED
STATUS_FLASHBACK
```

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```
---------- ------------------------ ------------------------ ---------- -------------- --------------
----------------
         1 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                       0              0              0
         2 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                       0              0              0
         3 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                       0              0              0
 exit
```

Name	Description
RP_NO	Number of the configured restore point – and it is the primary key of the table.
VALID_FROM	Timestamp from which the restore point can be created.
VALID_TO	Timestamp to which the restore point can be created.
SCN	Database SCN when the restore point was created
STATUS_CREATED	Create status: 0 = not created 1 = sucessfully created
STATUS_DROPPED	Drop status: 0 = not dropped 1 = sucessfully dropped
STATUS_FLASHBACK	Flashback status: 0 = not flashed back 1 = sucessfully flashed back

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1.3.1 Preconfigure Restore point through ADM Package

- Make sure all the old restore points are dropped before creating a new restore point. as concurrently only one restore point can be created in the database.
  select RP_NO, VALID_FROM, VALID_TO, SCN, STATUS_CREATED, STATUS_DROPPED, STATUS_FLASHBACK from adm.adm_restore_points;
- If status created is 1 (indicates rp created successfully) ,then status dropped should also be 1 (dropped successfully)
- If status dropped is 0 ,it means old restore point is not dropped yet.We have to drop it before creating a new one.
- **Preconfigure through admuser (script to be run from sql developer):**
- Change date value in the below script before running (todays date to date after one month)
- set serveroutput on
  declare
  t_rp_number NUMBER;
  begin
  t_rp_number := adm.p_adm.PreCfgRP(TO_DATE('2022-08-01', 'YYYY-MM-DD'), TO_DATE('2022-09-15', 'YYYY-MM-DD'), 'created using admin package') ;
  dbms_output.put_line (to_char (t_rp_number));
  end;
- Output : [0] PrecfgRP: Restore point 10004 has been pre-configured. 10004 is the preconfigured RP number,we can use it for simulation and creation.

1.3.2 Simulate Restore Point through ADM Package

- Scripts to check the status .Please check the status after each step by executing this script
  select * from adm.adm_restore_point_actvts where rp_no = 10004;
  select RP_NO, VALID_FROM, VALID_TO, SCN, STATUS_CREATED, STATUS_DROPPED, STATUS_FLASHBACK from adm.adm_restore_points;
- **Script to Simulate RP:**
- set serveroutput on
  begin
  adm.p_adm.simulaterp(p_rp_no => 10004);
  end;

1.4 Create a Restore Point

To create a guaranteed Restore Point procedure CreateRP of the p_adm packed has to be used.

Login in database via sqlplus as ADMUSER if not done and execute procedure p_adm.CreateRP(P_RP_NO).

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Then following steps are done:

- It waits for some seconds and restarts database instance.
- Session calling procedure CreateRP is lost.
- When restarting database instance the guaranteed restore point is set.
- Status of creation is stored in column STATUS_CREATED in table ADM_RESTORE_POINTS.
- Restore point is set asynchronously in a background job.

Check if creation was successful after restart of database provoked by execution of CreateRP.

- If creation was successful the value of STATUS_CREATED is 1.
- Values greater than 1 indicate an error. Details are specified in column STATUS_CREATED_TEXT.
- A Number is assigned to SCN.

Example:

Create Restore Point

```
d:\Puffer\oracle>sqlplus admuser@te4401.world

SQL*Plus: Release 11.2.0.2.0 Production on Do Feb 26 13:56:07 2015

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Oracle Database 11g Enterprise Edition Release 11.2.0.4.0 - 64bit Production
 set lines 132 pages 9999
 set serveroutput on
execute p_adm.createrp(1)
```

Check if Restore Point is successfully set.

```
d:\Puffer\oracle>sqlplus admuser@te4401.world

SQL*Plus: Release 11.2.0.2.0 Production on Do Feb 26 13:56:07 2015
```

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```
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Oracle Database 11g Enterprise Edition Release 11.2.0.4.0 - 64bit Production
 set lines 132 pages 9999
 column VALID_FROM format A25
 column VALID_TO format A25
  column SCN format 999999999999999
 set serveroutput on
 select RP_NO, VALID_FROM, VALID_TO, SCN, STATUS_CREATED, STATUS_DROPPED, STATUS_FLASHBACK  from adm.adm_restore_points;

  RP_NO VALID_FROM                VALID_TO                            SCN STATUS_CREATED STATUS_DROPPED
STATUS_FLASHBACK
------- ------------------------- ------------------------- ------------- -------------- --------------
----------------
      1 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000  7286967098652              1              0              0
      2 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                             0              0              0
      3 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                             0              0              0
 exit
```

1.5 Database Rollback

To roll back a database use procedure FlashbRP of the p_adm package.
To do this a valid Restore Point has to be created with procedure CreateRP.

Login in database via sqlplus as ADMUSER if not done and execute procedure p_adm.flashbrp(P_RP_NO) using the same P_RP_NO as used by creating the Restore Point.

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The following steps are done:

- It waits for some seconds and restarts database instance.
- Session calling procedure FlaschbRP is lost.
- When restarting database instance the database is rolled back to guaranteed restore point.
- Status of creation is stored in column STATUS_FLASHBACK in table ADM_RESTORE_POINTS.
- Flashback is done asynchronously in a background job.

Check if the rollback was successful after restart of database provoked by execution of FlashbRP

If flashback was successful the value is 1.

Another value > 1 indicates an error. Details are specified in column STATUS_FLASHBACK_TEXT.

Remark:

Restore point is not dropped by flashing back to it. It must be dropped if it is no longer needed.

Example:

Database roll back.

```
d:\Puffer\oracle>sqlplus admuser@te4401.world

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Kennwort eingeben:

Verbunden mit:
Oracle Database 11g Enterprise Edition Release 11.2.0.4.0 - 64bit Production
 set lines 132 pages 9999
 set serveroutput on
execute p_adm.flashbrp(1)
```

Check if roll back was successfully.

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```
d:\Puffer\oracle>sqlplus admuser@te4401.world

SQL*Plus: Release 11.2.0.2.0 Production on Do Feb 26 13:56:07 2015

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Kennwort eingeben:

Verbunden mit:
Oracle Database 11g Enterprise Edition Release 11.2.0.4.0 - 64bit Production
 set lines 132 pages 9999
 column VALID_FROM format A25
 column VALID_TO format A25
  column SCN format 999999999999999
 set serveroutput on
 select RP_NO, VALID_FROM, VALID_TO, SCN, STATUS_CREATED, STATUS_FLASHBACK  from adm.adm_restore_points;

  RP_NO VALID_FROM                VALID_TO                            SCN STATUS_CREATED STATUS_FLASHBACK
------- ------------------------- ------------------------- ------------ -------------- ----------------
      1 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000  7286967098652              1                1
      2 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                             0                0
      3 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                             0                0
 exit
```

1.6 Drop a Restore Point

To drop a restore point use procedure p_adm.DropRP.
After dropping the Restore Point all changes done after the Restore point can no longer rolled back.

Login in database via sqlplus as ADMUSER if not done and execute procedure p_adm.DropRP(P_RP_NO) using the same P_RP_NO as used by creating the Restore Point.

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Following steps are done:
Status of dropping is stored in column STATUS_DROPPED in ADM_RESTORE_POINTS.
Restore point is dropped asynchronously in a background job.
Check if the Restore Point is dropped successfully.
If dropping was successful the value is 1.
Another value > 1 indicates an error. Details are specified in column STATUS_DROPPED_TEXT.

Example:
Drop Restore Point.

```
d:\Puffer\oracle>sqlplus admuser@te4401.world

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Kennwort eingeben:

Verbunden mit:
Oracle Database 11g Enterprise Edition Release 11.2.0.4.0 - 64bit Production
 set lines 132 pages 9999
 set serveroutput on
execute p_adm.droprp(1)
```

Check if drop was successfully.

```
 column VALID_FROM format A25
 column VALID_TO format A25
  column SCN format 999999999999999
 set serveroutput on
 select RP_NO, VALID_FROM, VALID_TO, SCN, STATUS_CREATED, STATUS_DROPPED, from adm.adm_restore_points;

  RP_NO VALID_FROM                VALID_TO                            SCN STATUS_CREATED STATUS_DROPPED
```

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```
------- ------------------------ ------------------------ ------------- -------------- --------------
      1 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000  7286967098652              1              1
      2 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                             0              0
      3 28.02.15 00:00:00,000000  02.03.15 00:00:00,000000                             0              0
 exit
```

1.7 Check for Restore Point set by database maintainer outside P_ADM package.

To check if there are Restore Points outside the P:_ADM package login as user **INFODBA** and execute the sql statements shown below.

```
 column CURRENT_SCN format 999999999999999
 column name format a25
 column TIME format A25
 SELECT NAME, SCN, to_Char(TIME, 'YYYY-MM-DD HH24:MM-SS') TIME, DATABASE_INCARNATION#,
GUARANTEE_FLASHBACK_DATABASE,STORAGE_SIZE FROM V$RESTORE_POINT;
```

```
NAME                      SCN TIME                DATABASE_INCARNATION# GUA STORAGE_SIZE
------------------------- ------------- ------------------------ --------------------- --- ------------
BEFORE_TE4303          7414035157705  2015-11-27 14:11-40                    2 YES   1,2688E+10
```

To check if a Restore Point is set and whether Flashback is active or not check database Parameter FLASHBACK_ON.

```
 select FLASHBACK_ON from v$database;
FLASHBACK_ON
------------------
NO
```

1.8 Check the size of the FLASHBACK_AREA

To check the size of the FLASHBACK_AREA and the size used by RESTORE_POINTS login as user **INFODBA** and execute the sql statements shown below.

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```
column NAME format A45
SELECT NAME,
      ROUND(SPACE_LIMIT / 1048576) SPACE_LIMIT_MB,
      ROUND(SPACE_USED / 1048576) SPACE_USED_MB,
      ROUND(((SPACE_USED / 1048576) * 100) / (SPACE_LIMIT / 1048576), 2) PRC_USED
FROM V$RECOVERY_FILE_DEST;

NAME                                          SPACE_LIMIT_MB SPACE_USED_MB   PRC_USED
--------------------------------------------- -------------- ------------- ----------
/oradata4_cl-000l/TE4801/flashback                      5120           600      11,72
```

with

`Name`	`S` torage area
`SPACE_LIMIT_MB`	Max size of storage area in MB
`SPACE_USED_MB`	Size used by Restore Points in MB
`PRC_USED`	Percentage of storage area used by Restore Points

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2 Database Flashback (CI/DAP54.2 T/Q Databases Only)

WARNING: You will use this manual at your own risk. Wrong handling may crash the database.

This manual is only valid for CI/DAP52.4 self maintained Oracle databases hosted on database server SI0BOS419.

Flashback is an easy workaround for setting back changes or recover deleted data.
To revert the entire database to a prior point in time, you can either revert the entire database to a prior point in time by restoring a backup and doing point-in-time recovery, or you can enable Flashback Database. When you enable Flashback Database, the database generates Flashback logs in the fast recovery area. These logs are used to flash back the database to a specified time. During usual operation, the database occasionally logs images of data blocks to the Flashback logs. The database automatically creates, deletes, and resizes Flashback logs.
In the following shows how to revert changes in a database using FLASHBACK DATABASE.

Connect to TC T/Q Oracle Database Server (si0bos419).
Set environment variable ORACLE_SID to the approbriate database.

C:\Users\oraiman> `set ORACLE_SID=te####`

C:\Users\oraiman> `sqlplus /nolog`

SQL*Plus: Release 12.1.0.2.0 Production on Di Sep 3 15:37:58 2019
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SQL> `connect / as sysdba`

Connect durchgef³hrt.
SQL>

2.1 Prerequisite of Flashback Database

To use the Flashback functionally it has to be ensured that a fast recovery area is configured and the database is running in **ARCHIVELOG** mode.
To check if you are connected to the right database and the database is running in ARCHIVELOG mode log execute

```
SELECT name, log_mode , flashback_on  FROM v$database;
```

Sample:

```
SQL> SELECT name, flashback_on, log_mode FROM v$database;

NAME         LOG_MODE     FLASHBACK_ON
------------ ------------ ------------
TE3001       ARCHIVELOG   NO
```

For relocation the fast recovery area and to define/adjust its size following initialization parameters are used:

```
DB_RECOVERY_FILE_DEST
```

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Specifies the location of the fast recovery area.

`DB_RECOVERY_FILE_DEST_SIZE`

Specifies the size of the fast recovery area, in bytes.

Hint:

If the database is not running in ARCHIVELOG mode and/or the fast recovery area isn't configured you may stop going on reading and think about another methods to roll back a database to a prior date after doing some changes.

2.2 Enable Flashback Database

To use the Flashback functionality Flashback mode has to be enabled.
Log in to the database and use the following command to check if Flashback Database is enabled

```
SELECT name, flashback_on, log_mode FROM v$database;
```

Sample:

```
SQL> SELECT name, flashback_on, log_mode FROM v$database;
NAME         LOG_MODE     FLASHBACK_ON
------------ ------------ ------------
TE3001       ARCHIVELOG   NO
```

If Flashback mode is `enabled (FLASHBACK_ON=YES)` goto Set a GUARANTEED RESTORE POINT

If Flashback mode is `disabled ( FLASHBACK_ON =NO)` execute following steps
Shutdown the database instance. It is recommended to stop all processes/services connected to the database before.

```
shutdown immediate;
```

Startup the instance and mount the database without opening it.

```
startup mount;
```

Enable the Flashback Database feature for the whole database.

```
alter database flashback on;
```

Open the database for general use.

```
alter database open;
```

Sample:

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```
SQL> SELECT name, flashback_on, log_mode FROM v$database;
NAME      FLASHBACK_ON       LOG_MODE
--------- ------------------ ------------
TE3401    NO                 ARCHIVELOG

SQL> shutdown immediate;
Datenbank geschlossen.
Datenbank dismounted.
ORACLE-Instanz heruntergefahren.
SQL> startup mount;
ORACLE-Instanz hochgefahren.
Total System Global Area 2634022912 bytes
Fixed Size                  3835544 bytes
Variable Size             738199912 bytes
Database Buffers         1879048192 bytes
Redo Buffers               12939264 bytes
Datenbank mounted.
SQL>
SQL> alter database flashback on;
Datenbank wurde geõndert.
SQL>
SQL> alter database open;
Datenbank wurde geõndert.
SQL>
SQL> SELECT name, flashback_on, log_mode FROM v$database;
NAME      FLASHBACK_ON       LOG_MODE
--------- ------------------ ------------
TE3401    YES                ARCHIVELOG
```

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```
SQL>
```

2.3 Set a GUARANTEED RESTORE POINT

If Flashback mode is enabled Restore Points can be created for future use to roll back the database.
For this a GUARANTEED RESTORE POINT has to be created.
Use following statement like

```
create restore point <RP_NAME> GUARANTEE Flashback DATABASE;
```

Example:

```
create restore point BEFORE_TC11606 GUARANTEE Flashback DATABASE;
```

Check for Restore Points set:

```
column NAME format a25
column TIME format A25
column SCN format 999999999999999
column STORAGE_SIZE format 999999999999
SELECT NAME, SCN, to_Char(TIME, 'YYYY-MM-DD HH24:MM:SS') TIME, GUARANTEE_Flashback_DATABASE,STORAGE_SIZE FROM
V$RESTORE_POINT;
```

Sample:

```
SQL> column NAME format a25
 SQL>  column TIME format A25
 SQL>  column SCN format 999999999999999
 SQL>  column STORAGE_SIZE format 999999999999
 SQL>  SELECT NAME, SCN, to_Char(TIME, 'YYYY-MM-DD HH24:MM:SS') TIME, GUARANTEE_Flashback_DATABASE,STORAGE_SIZE FROM
V$RESTORE_POINT;
NAME            SCN            TIME                GUA     STORAGE_SIZE
--------------- -------------- ------------------- --- ----------------
BEFORE_TC11606    7415279875248 2019-08-22 06:08:59  YES       1048576000
```

Hint: A guaranteed Restore Point set can be used several times to flashback the database.

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2.4 Perform a Flashback Database Operation

This section shows how to perform a flashback of the database, specifying the name of a normal or guaranteed restore point, or an SCN.
Check for available guaranteed Restore Points

```
column NAME format a25
column TIME format A25
column SCN format 999999999999999
colum STORAGE_SIZE format 999999999999
SELECT NAME, SCN, to_Char(TIME, 'YYYY-MM-DD HH24:MM:SS') TIME, GUARANTEE_Flashback_DATABASE,STORAGE_SIZE FROM
V$RESTORE_POINT;
```

Shutdown the database instance. It is recommended to stop all processes/services connected to the database before.

```
shutdown immediate;
```

Startup the instance and mount the database without opening it.

```
startup mount;
```

Now you can flashback the database by using a SCN

```
flashback database to SCN  <SCN_No.>;
```

like

```
flashback database to SCN  3 5750578 ;
```

or a Restore Point name

```
flashback database to RESTORE POINT <RP_NAME>;
```

like

```
flashback database to RESTORE POINT  BEFORE_TC11602 ;
```

Open database with RESETLOGS option

```
alter database open RESETLOGS;
```

Now the database is open for general use.

Sample:

```
SQL>
SQL> -- existing restore points
SQL>  column NAME format a25
```

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```
SQL> column TIME format A25
SQL> column SCN format 999999999999999
SQL> column STORAGE_SIZE format 999999999999
SQL> SELECT NAME, SCN, to_Char(TIME, 'YYYY-MM-DD HH24:MM:SS') TIME, GUARANTEE_Flashback_DATABASE,STORAGE_SIZE FROM
V$RESTORE_POINT;
NAME                      SCN TIME                 GUA     STORAGE_SIZE
---------------  ------------ -------------------- --- ----------------
BEFORE_TC11602        35750578  2019-07-25 13:07:42  YES       1048576000

-- DB zurückrollen auf Stand BEFORE_TC11602
SQL>
SQL> shutdown immediate;
Datenbank geschlossen.
Datenbank dismounted.
ORACLE-Instanz heruntergefahren.
SQL> startup mount;
ORACLE-Instanz hochgefahren.
Total System Global Area 4731174912 bytes
Fixed Size                  5414424 bytes
Variable Size            1073742312 bytes
Database Buffers         3640655872 bytes
Redo Buffers               11362304 bytes
Datenbank mounted.
SQL> flashback database to RESTORE POINT BEFORE_TC11602;
Flashback abgeschlossen.
SQL> alter database open resetlogs;
Datenbank wurde geõndert.
SQL>
```

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2.5 Drop Restore Point

When a Restore Point is no longer needed for a roll back it should become dropped.
Without dropping a restore point the application will no longer be able to do insert/update/deletes when the Flashback area is full.
So it is important to drop the restore point when the maintenance is over.
To drop a Restore Pint execute following statement:

```
drop restore point <RP_NAME>;
```

Sample:

```
SQL> drop restore point BEFORE_TC11606;
Restore-Punkt gelöscht.
SQL>
```

2.6 Disable Flashback

To disable Flashback

```
alter database Flashback off;
```

After disabling Flashback the claimed storage space on file system will be released.

Example:

```
SQL> alter database flashback off;
Datenbank wurde geõndert.
SQL>
```

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